

Education

• M.Sc. Data Science, Monash University, Australia.	2024–2026
• B.Sc. (Hons.) Computer Science, University of Nottingham, UK.	2019–2020
• B.Sc. Computer Science, Bina Nusantara International, Indonesia.	2016–2020

Work Experience

Associate Data Scientist	tiket.com	Oct 2022–July 2024
Jakarta, Indonesia		

tiket.com is one of Indonesia’s largest online travel agency companies, offering an app-based platform for booking flights, hotels, trains, car rentals, and entertainment tickets.

- Increased hotel booking conversion rate by 2.3% (IDR 47B annual GBV impact) by deploying a deep learning image quality classifier that automatically filtered low-quality hotel photos, reducing bounce rate from poor listings by 18% across 15,000+ properties.
- Increased payment conversion by 4.8% through rigorous A/B experimentation of a Scikit-learn recommendation engine, analyzing user behavior via Amplitude event tracking across 15+ payment methods and 4 user segments, resulting in IDR 94B incremental GBV and informing company-wide payment gateway optimization strategy.
- Reduced duplicate hotel room listings by 34% (improving search relevance for 8M monthly users) by building a CatBoost similarity model that automatically grouped identical rooms across 12,000 properties, cutting manual review time from 120 to 15 hours/week.
- Drove IDR 8B incremental GBV by A/B testing ensemble model improvements to the hotel recommendation engine, conducting segmented analysis across domestic vs. international to uncover that POI proximity features delivered 18% higher insights that shaped personalization roadmap and deployment prioritization across major SEA markets.
- Saved IDR 315M annually in operational costs by deploying an LSTM-based flight disruption prediction system that identified 82% of reschedules 4+ hours in advance, enabling proactive customer communication and reducing call center volume by 23%.
- Built executive dashboards in Tableau tracking 12 KPIs across flight/hotel verticals, enabling weekly data-driven decisions for product team members.

Projects & Achievements

Causal Inference & Uplift Modeling	Personal	Sep 2025-Oct 2025
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- Implementing and comparing causal inference methods (Propensity Score Matching, Double Machine Learning, uplift modeling) to estimate heterogeneous treatment effects from observational data, addressing common A/B testing limitations when randomization is infeasible.
- Built an end-to-end pipeline using Python (EconML, CausalML libraries) to estimate average treatment effects and conditional average treatment effects (CATE), demonstrating 80-90% improvement in treatment effect estimation precision compared to naive observational comparisons.
- Developed a dashboard using Streamlit to visualize the key metrics, method comparison, and interactive controls with live updates.

Slot Filling & Intent Detection	Monash University	Sep 2025-Oct 2025
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- Deployed all four models to a Streamlit environment, with a 90% accuracy that simultaneously predicts from 21 unique intents and 121 unique slot labels using shared representations.
- Performed multiple model comparisons from Baseline (CRF + Random Forest) to state-of-the-art models like JointBiLSTM, JointBiLSTM-Attention, and JointBERT with +95% intent accuracy and +80% slot F1-score.

Technologies and Languages

• Languages	: Python, R, C++, SQL
• ML/DL Frameworks	: PyTorch, TensorFlow, Scikit-learn, XGBoost, CatBoost
• Data Engineering	: Apache Spark (PySpark), Kafka, BigQuery, MongoDB
• Experimentation & Statistics	: A/B Testing, Hypothesis Testing
• Cloud & MLOps	: GCP (Vertex AI, Cloud Functions), Docker
• Analytics & Visualization	: Tableau, D3.js, Matplotlib, Seaborn
• Developer Tools	: Git, Jupyter Notebook, CI/CD Pipelines